

Weighted Graph-Based Event Clustering and Priority Scoring for Flood-Rescue Coordination Using Edge AI

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Abstract. During flood disasters, telecommunication infrastructure often collapses, disabling cloud-centric processing exactly during the rescue “golden hour.” We propose an end-to-end framework combining Edge AI and weighted graph theory to collect, cluster, and automatically rank rescue events. Edge devices extract a multidimensional attribute vector (L, T, F, E, N, V, C) from images and text, then transmit only a sub-kilobyte metadata packet instead of raw media. On the server, events are modeled as a weighted graph in which geographic distance acts as a *multiplicative gate* rather than an additive term, guaranteeing that every cluster is spatially coherent. The Louvain algorithm (with Leiden recommended) decomposes the graph into “operational zones,” and a cluster-level priority function $\mathcal{P}(C_k)$ —built from a normalized risk core and a demographic-vulnerability factor acting as an *amplifier*—ranks clusters to support dispatch. On a deterministic synthetic dataset of 341 events modeled on Central Vietnam, the gating form cuts the *worst-case* cluster diameter from **214 km to 1.4 km** while *also* improving clustering accuracy (Adjusted Rand Index, ARI $0.957 \rightarrow 0.996$) over the strongest additive configuration found by an α sweep; across 20 seeds the gating form wins on 100% of seeds on every metric (ARI 0.9957 ± 0.0000 vs. 0.9415 ± 0.0141). A confidence gate C_i suppresses fake reports that inflate victim counts (a **55%** reduction of phantom population), though an adversary combining a forged image with staged corroboration reaches $C_i = 0.92$ —indistinguishable from real reports, a limitation we report rather than hide. Against baselines on the *same* gating graph, Spectral clustering collapses (0.166) and HDBSCAN attains a perfect ARI but with a 55.7 km mean cluster diameter that is operationally unusable, while K-Means (0.502) and DBSCAN (0.523) on raw coordinates trail badly; priority rankings are stable (Kendall’s τ : mean 0.955, min 0.910 under ± 0.10 weight perturbation, top-3 preserved in 100% of trials).

Keywords: Edge AI · Event clustering · Weighted graph · Community detection · Louvain · Rescue prioritization · Multimodal analysis · Flood disaster.

1 Introduction

Climate change is intensifying the frequency and severity of extreme weather events. Vietnam—with its long coastline and terrain directly exposed to typhoon circulation—is affected by roughly 6–8 typhoons and tropical cyclones each year (out of the ~ 11 storms and tropical depressions that develop in the East Sea) [11], causing heavy losses of life and property, especially in the central and northern regions.

A core cause of failure in emergency response is the disruption of telecommunication infrastructure. When the power grid degrades and base transceiver stations (BTS) become isolated, cloud-centric data collection and processing pipelines fail entirely, cutting command centers off from the storm’s epicenter precisely during the “golden hour.” In this context, social media and messaging applications become a survival-critical *social sensing* channel, producing a *multimodal* data stream (text, images, video, spatiotemporal metadata) that is fragmented, duplicated, noisy, and prone to *information overload* [10].

This paper proposes an end-to-end framework that simultaneously addresses three challenges:

1. **Surviving degraded networks:** push AI to edge devices; instead of uploading megabytes of images/video, the application processes on-device and transmits only a sub-kilobyte metadata packet.
2. **Physically meaningful de-duplication of events:** build a weighted graph integrating space, time, and hazard context, then perform community detection.
3. **Automatic priority ranking:** quantify urgency at the *cluster* level to answer the dispatch question “which cluster first?”

Main contributions.

- A **multiplicative (gating) edge-weight function** in which geographic similarity globally modulates non-spatial similarity, guaranteeing spatially coherent clusters—a prerequisite for dispatching boats with a finite operating radius.
- A **cluster-level priority function** $\mathcal{P}(C_k)$ with a normalized risk core and a **demographic-vulnerability factor as a multiplicative amplifier**, injecting equity into the resource-allocation problem.
- Two **edge-feasible** auxiliary attributes: a vulnerability index V_i (folded into the existing text classifier) and a confidence score C_i (a lightweight sigmoid heuristic), both providing robustness against fake reports.
- A comprehensive **empirical validation** on a deterministic synthetic dataset modeled on Central Vietnam, justifying each design decision with quantitative evidence.

2 Related Work

2.1 Multimodal Analysis in Crisis

The shift from unimodal to multimodal learning has substantially improved the situational awareness of disaster-management systems. Pioneering datasets such as CrisisMMD [1] and FloodNet [17] provide training foundations for damage assessment and urgency classification. For images/video, convolutional networks such as ResNet and MobileNetV3 [8], or semantic-segmentation models, are used to extract flooded regions; some work applies human pose estimation to infer water depth where no physical gauges exist. For text/audio, Transformer architectures (BERT, DistilBERT [20]) perform sentiment analysis, entity recognition, and needs identification. Graph-neural-network approaches such as CrisisSpot [4] show that fusing modalities improves disaster-content classification F1 by 5.01–9.45%.

2.2 Edge Computing for Disaster Response

The critical bottleneck of multimodal models is their bandwidth and compute demand. During floods, uploading high-resolution media to the cloud is infeasible. The community therefore promotes *Edge AI* [13,15]: model compression (quantization, knowledge distillation) and lightweight architectures enable on-device inference. The edge device transmits only a compact metadata packet (sub-kilobyte, quantified in Sect. 6), so distress signals still penetrate congested networks. EmergencyNet [13] exemplifies a lightweight CNN for drone/embedded platforms with millisecond-scale inference latency.

2.3 Graph-Based Spatiotemporal Analytics

Treating each distress call as an isolated point discards global context. Graph theory offers a mathematical framework to model interactions and risk propagation: a rescue event is a vertex v in a graph $G = (V, E, W)$, with edges E encoding relationships and weights W their strength. Event-detection studies typically build weighted graphs from spatiotemporal proximity and keyword co-occurrence (e.g., TF-IDF similarity). Advanced models incorporate Euclidean/Haversine distance to *penalize* links between distant events, forming geo-semantic graphs—yet they still omit flood depth and demographic vulnerability.

2.4 Community Detection

Traditional unsupervised methods are limited: **K-Means** [14] requires a pre-defined cluster count K (infeasible in a volatile disaster) and assumes spherical clusters; **DBSCAN** [5] is sensitive to high dimensionality and parameters. Network-topology methods, notably **Louvain** [2], are the gold standard for weighted-graph clustering via optimization of the **Modularity** function [16], with observed near- $\mathcal{O}(N \log N)$ runtime. The **Leiden** [21] variant fixes the badly-connected-community pathology that Louvain occasionally exhibits. A broader survey of the field is given by Fortunato [6].

Table 1. Positioning against representative work (✓ yes, ∼ partial, blank no).

Study	Multimodal	Edge/on-device	Weighted graph	Vulnerability	Cluster priority
CrisisSpot [4]	✓		✓ (GNN)		
Event detection (TF-IDF)			✓		
EmergencyNet [13]		✓			
Vulnerability prioritization [7]				✓	∼
Proposed framework	✓	✓	✓ (gating)	✓ (factor)	✓

2.5 Positioning

Table 1 positions this work against representative studies on five capability axes. No prior work simultaneously (i) runs multimodal extraction at the edge, (ii) encodes it into a spatial–semantic–physical gating weighted graph, (iii) integrates demographic vulnerability as an amplification factor, and (iv) produces a cluster-level priority ranking. That empty cell is the gap this paper fills.

3 Research Gaps

A literature review reveals three fundamental gaps, motivating the proposed framework.

Gap 1 — Missing hazard-specific attributes in edge weighting. Two geographically close events do not necessarily share the same risk, which is also governed by micro-topography, building structure, and victim condition. Omitting flood depth (from images) and urgency (from text) from the edge weight prevents the graph from distinguishing a group trapped on a rooftop from a group safe on a high floor at the same coordinates.

Gap 2 — Vulnerability blind spot. Humanitarian-logistics frameworks often assume homogeneous demand and minimize only travel distance/time [22]. But disasters impact groups unequally: the elderly, children, pregnant women, and the disabled deteriorate faster. Failing to *amplify* priority for events containing vulnerable individuals is a serious ethical (equity) shortcoming [7].

Gap 3 — Static clustering, no community-level priority. Many studies stop at “detecting event groups.” But coordinators need to know which cluster to dispatch to *first*. A cluster-level priority mechanism—combining maximum flood depth, total trapped count, vulnerable fraction, and overall urgency—has not been rigorously formalized.

4 Proposed Framework

The framework has four sequential stages (Fig. 1): (4.1) edge extraction of a multidimensional attribute vector; (4.2) construction of a spatial–semantic–physical weighted graph; (4.3) community decomposition via Louvain/Leiden; (4.4) cluster-level priority scoring. Stage 1 runs on the reporter’s device; stages

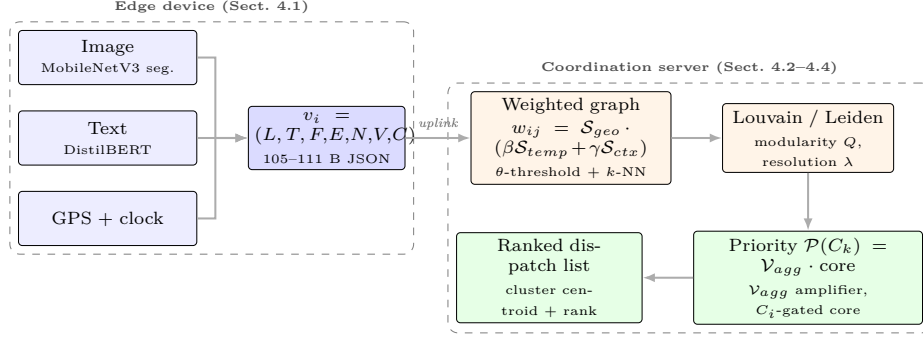


Fig. 1. End-to-end architecture. The edge device compresses raw multimedia into a 105–111-byte attribute vector; the server builds the gating weighted graph, decomposes it into operational zones, and ranks them. Only the metadata packet crosses the degraded link.

2–4 run on a coordination server that receives only metadata packets, so the whole pipeline survives a link that can carry kilobytes but not megabytes.

4.1 Multidimensional Attribute Vector

Each rescue event v_i is represented by seven attributes:

$$v_i = (L_i, T_i, F_i, E_i, N_i, V_i, C_i) \quad (1)$$

where L_i is GPS location, T_i a timestamp, $F_i \in [0, 1]$ physical flood level (MobileNetV3 segmentation/pose estimation), $E_i \in [0, 1]$ urgency (DistilBERT/UIT-VSMEC text sentiment), $N_i \in \mathbb{Z}^+$ the trapped count, $V_i \geq 0$ a demographic-vulnerability index, and $C_i \in (0, 1)$ an information-confidence score. The edge device transmits only a JSON string of these values (sub-kilobyte; Sect. 6) instead of megabytes of raw media.

Edge feasibility. The base attributes L, T, E, N, F follow the project’s commitments (images via MobileNetV3, quantized DistilBERT for text). The two auxiliary attributes are designed to add *no* heavy deep model: V_i is extracted by a multi-label head sharing the same text classifier used for E_i (detecting phrases such as “infant present,” “exhausted elderly,” “pregnant woman”), and C_i uses a lightweight sigmoid heuristic. Full versions (dedicated crowd counting/pose estimation for N_i ; a confidence model learned from user history for C_i) are explicitly positioned as future work beyond the six-month prototype.

Operational definition of V_i . V_i is the sum of humanitarian-priority weights over the vulnerable labels detected in a report. Table 2 gives the *proposed* weight scheme and its mapping to the four discrete levels $\{0, 1, 1.5, 2\}$ used in the

Table 2. Proposed vulnerability label weights and their mapping to the dataset’s four discrete V_i levels.

Detected label(s)	V_i level	Rationale
None	0.0	No vulnerable individual reported
Elderly <i>or</i> disabled (one)	1.0	Single standard vulnerable individual
Pregnant woman, or two labels	1.5	Elevated medical urgency / compounding
Infant, or three+ labels	2.0	Highest deterioration rate

dataset. We stress a scope boundary: the experiments assume V_i is *given* (ground-truth vulnerability), so they test the amplification *mechanism* of $\mathcal{V}_{agg}$, *not* the capability to extract V_i from raw text—the latter is future work. The weight table is a normative proposal (who sets it is a policy decision for command staff), not an empirically optimized artifact.

Confidence heuristic.

$$C_i = \sigma(b_0 + b_1 \cdot \mathbb{K}[\text{has image}] + b_2 \cdot \log(1 + n_i^{\text{corrob}})) \quad (2)$$

Here σ is the sigmoid, $\mathbb{K}[\text{has image}]$ indicates vision-verified multimodal evidence, and n_i^{corrob} counts independent corroborating reports within a corroboration radius $r_{\text{corrob}} = 400$ m and time window $\Delta t_{\text{corrob}} = 60$ min (kept deliberately separate from the clustering constants $\sigma_{geo}, \tau_{temp}$ to avoid cross-coupling). Log compression makes the 2nd–3rd corroborating report boost confidence strongly while the 50th adds almost nothing, resisting single-location spam. In experiments we use $(b_0, b_1, b_2) = (-0.2, 1.4, 0.9)$. Absent a persistent-account infrastructure, “independence” is approximated purely by spatiotemporal proximity; we return to the resulting vulnerability to coordinated fake corroboration in the adversarial analysis of Sect. 5.9.

4.2 Spatial–Semantic–Physical Weighted Graph

This is the most important design fix relative to a naive draft. A *naive additive* weighting

$$w_{ij} = \alpha \mathcal{S}_{geo} + \beta \mathcal{S}_{temp} + \gamma \mathcal{S}_{context} \quad (3)$$

fails: two events 50 km apart ($\mathcal{S}_{geo} \approx 0$) but both describing “water up to the roof” ($\mathcal{S}_{context} \approx 1$) still receive a substantial $w_{ij} \approx \gamma$. The clustering algorithm would then merge distant points into one “operational zone”—meaningless for a boat with a finite radius.

We instead adopt a **multiplicative (gating)** form in which $\mathcal{S}_{geo}$ sits *outside* as a global modulating factor:

$$w_{ij} = \mathcal{S}_{geo}(L_i, L_j) \cdot \left(\beta \cdot \mathcal{S}_{temp}(T_i, T_j) + \gamma \cdot \mathcal{S}_{context}(v_i, v_j) \right) \quad (4)$$

When distance is large, $\mathcal{S}_{geo} \rightarrow 0$ forces $w_{ij} \rightarrow 0$ regardless of context similarity—geography dominates cluster structure. The components are:

$$\mathcal{S}_{geo} = \exp\left(-\frac{\text{dist}(L_i, L_j)^2}{2\sigma_{geo}^2}\right) \quad (5)$$

$$\mathcal{S}_{temp} = \exp\left(-\frac{|T_i - T_j|}{\tau_{temp}}\right) \quad (6)$$

$$\mathcal{S}_{context} = \exp\left(-\frac{|F_i - F_j|}{\tau_F} - \frac{|E_i - E_j|}{\tau_E}\right) \quad (7)$$

$\text{dist}(\cdot)$ is Haversine distance (meters); σ_{geo} is a characteristic radius approximating a boat’s operating range (a few hundred meters to 1–2 km). The squared distance makes the Gaussian gate decay sharply, strongly penalizing distant links. The temporal kernel uses a first-order (non-squared) exponent because flood evolution has inertia. For context, similar reports ($\Delta F \approx 0, \Delta E \approx 0$) give $\mathcal{S}_{context} \rightarrow 1$, whereas dissimilar ones (one safe on a 3rd floor, one clinging to a rooftop) shrink it—correctly reflecting distinct rescue needs. Since $\exp(-a - b) = \exp(-a)\exp(-b)$, both conditions (similar flood *and* similar urgency) must hold jointly for a high $\mathcal{S}_{context}$. The looser $\tau_E = 0.35$ versus $\tau_F = 0.25$ makes the kernel more tolerant of a given urgency gap than of the same flood-depth gap, reflecting that the sentiment-derived E is noisier than the vision-derived F and therefore warrants a wider matching band; a sensitivity sweep (Sect. 5.3) confirms clustering is insensitive to both within a wide band, so this asymmetry is a mild modeling preference rather than a tuned parameter. The additive form’s α is eliminated because $\mathcal{S}_{geo}$ is now a global factor.

Graph sparsification. Because $\mathcal{S}_{geo}$ decays quickly, most long-range edges carry negligible weight. We sparsify via either (i) a threshold θ (retain edges with $w_{ij} > \theta$) or (ii) a k -NN graph (each vertex keeps its k highest-weight neighbors). This reduces computation and removes spurious cross-zone links, since Modularity optimization degrades on dense, near-complete graphs.

4.3 Clustering via Louvain (Leiden Recommended)

We partition the vertex set into disjoint clusters $\{C_1, \dots, C_k\}$ maximizing Modularity in the Reichardt–Bornholdt [18] resolution form:

$$Q = \frac{1}{2m} \sum_{i,j} \left[A_{ij} - \lambda \frac{k_i k_j}{2m} \right] \delta(c_i, c_j) \quad (8)$$

where $A_{ij} = w_{ij}$, $k_i = \sum_j A_{ij}$ is the weighted degree, $m = \frac{1}{2} \sum_{i,j} A_{ij}$ is total graph weight, δ is the Kronecker delta, and $\frac{k_i k_j}{2m}$ is the null-model expected edge weight. The resolution λ controls granularity: $\lambda = 1$ is standard Modularity; $\lambda > 1$ splits more finely (a large flooded ward into distinct blocks); $\lambda < 1$ favors larger clusters.

Louvain is chosen because it (1) auto-discovers the cluster count—no a-priori K ; (2) denoises structurally—fake reports lack strong edges and become singletons; (3) has observed near- $\mathcal{O}(N \log N)$ runtime for real-time use; and (4) is student-feasible via `python-louvain/networkx/igraph`. We recommend the **Leiden** [21] refinement, which guarantees well-connected communities, since a badly connected cluster would place the dispatch centroid away from the actual victims.

4.4 Cluster-Level Priority Scoring

Identifying clusters is insufficient: which cluster gets the first boat? We propose a priority function that fixes three mathematical flaws of a naive additive score: (a) scale bias, (b) vulnerability modeled as an additive rather than multiplicative term, and (c) early tanh saturation.

$$\mathcal{P}(C_k) = \mathcal{V}_{agg}(C_k) \cdot \left(\omega_1 \tilde{\mathcal{E}}_{agg}(C_k) + \omega_2 \tilde{\mathcal{F}}_{max}(C_k) + \omega_3 \tilde{\mathcal{N}}(C_k) \right) \quad (9)$$

(a) *Scale normalization.* $\mathcal{E}_{agg}, \mathcal{F}_{max} \in [0, 1]$ but $\mathcal{N}_{total} = \sum N_i$ is unbounded (possibly hundreds). Adding them directly lets population dominate. We normalize every component to $[0, 1]$ before weighting $(\tilde{\cdot})$. For the right-skewed population, we log-compress then min-max:

$$\tilde{\mathcal{N}}(C_k) = \frac{\log(1 + \mathcal{N}_{total}(C_k))}{\log(1 + N_{\max})} \quad (10)$$

where $N_{\max}$ is a reference. A *dynamic* reference (largest current-window cluster) gives instantaneous relative ranking; a *fixed* reference is needed for across-time comparison. The dynamic mode is non-stationary and must be interpreted accordingly.

Risk-core components (all confidence-gated).

$$\mathcal{E}_{agg}(C_k) = \frac{1}{|C_k|} \sum_{v_i \in C_k} E_i \cdot C_i \quad (11)$$

$$\mathcal{F}_{max}(C_k) = \max_{v_i \in C_k} (F_i \cdot C_i) \quad (12)$$

$$\mathcal{N}_{total}(C_k) = \sum_{v_i \in C_k} N_i \cdot C_i \quad (13)$$

We use max (not mean) for flood depth by the “communicating-vessels” principle: the deepest point determines the cluster’s survival risk; averaging would dilute the alarm. Crucially, C_i is multiplied *inside* the max so a low-confidence fake report claiming $F = 1.0$ cannot capture $\mathcal{F}_{max}$ —restoring consistency with the confidence-gating of $\mathcal{E}$ and $\mathcal{N}$.

(b,c) *Vulnerability amplification.*

$$\mathcal{V}_{agg}(C_k) = 1 + \tanh\left(\frac{1}{s} \sum_{v_i \in C_k} V_i\right), \quad \mathcal{V}_{agg} \in [1, 2] \quad (14)$$

Placing $\mathcal{V}_{agg}$ *outside* as a factor makes it a true amplifier: a cluster with no vulnerable individuals gets ≈ 1 (risk core unchanged), while one with many gets up to $2\times$. An additive term bounded in $[1, 2]$ would only add a near-constant offset—amplifying nothing. The scale factor s (e.g., $s = 10$) prevents early saturation: with raw $\tanh(\sum V_i)$, only 2–3 vulnerable individuals push $\tanh$ near 1, making a cluster with 1 indistinguishable from one with 50; dividing by s stretches the linear region while $\tanh$ still caps the value. An equivalent alternative is $1 + \log(1 + \sum V_i)$ with normalization.

Adjustable amplification cap. The fixed $2\times$ ceiling is a policy choice, not a mathematical necessity. To let command staff bound how far vulnerability may reorder the list—a genuine “many-healthy vs. few-vulnerable” trade-off—we generalize to

$$\mathcal{V}_{agg}(C_k) = 1 + (\mu - 1) \tanh\left(\frac{1}{s} \sum_{v_i \in C_k} V_i\right), \quad \mu \in [1, 2], \quad (15)$$

so $\mu = 1$ disables amplification (pure risk core) and $\mu = 2$ recovers the default. All reported numbers use $\mu = 2$. A larger μ can push a small vulnerable-rich cluster above a large healthy one; exposing μ makes that ethical dial explicit rather than hard-coded.

Weights and range. $\omega_1, \omega_2, \omega_3$ (with $\sum \omega = 1$) are set by command staff via an Analytic Hierarchy Process (AHP) decision matrix [19] to switch tactical modes (population-first vs. depth-first). Since the core is normalized to $[0, 1]$ and $\sum \omega = 1$, the core lies in $[0, 1]$; multiplying by $\mathcal{V}_{agg} \in [1, 2]$ bounds $\mathcal{P}(C_k) \in [0, 2]$ —convenient for ranking. Descending $\mathcal{P}(C_k)$ yields an actionable priority list; combined with cluster centroids it is an ideal input for downstream route optimization (A^* cost-aware, multi-commodity routing).

On reusing F, E in two stages. F and E appear both in clustering (via $\mathcal{S}_{context}$) and prioritization (via $\mathcal{F}_{max}, \mathcal{E}_{agg}$). This is not erroneous double-counting: $\mathcal{S}_{context}$ measures pairwise *similarity* (whether two events share a situation), whereas $\mathcal{F}_{max}/\mathcal{E}_{agg}$ measure a cluster’s *absolute severity* (for ranking). We acknowledge, however, that a cluster grouped by similar F will almost certainly have high $\mathcal{F}_{max}$, so $\mathcal{F}_{max}$ should be read as “the characteristic flood level of an already-homogeneous population,” not a fully independent signal. We quantify this circularity in Sect. 5.7: removing $\mathcal{S}_{context}$ from the graph leaves the priority ranking *bit-identical* (Kendall’s $\tau = 1.0$), so the context term cannot be covertly dictating the *ranking*; where it does matter is separating groups closer together than σ_{geo} , which the β/γ sweep of Sect. 5.3 isolates.

5 Experiments

5.1 Setup

Dataset. Absent a real labeled ground-truth dataset for rescue clusters, we generate a *deterministic synthetic* dataset (seed = 42) modeled on Central Vietnam (Huế–Quảng Trị–Quảng Nam–Đà Nẵng; 15.7–17.1°N, 107.0–108.6°E) with **341 events** spanning **14 ground-truth labels**: 240 core events around **6 flood “islands”** (40 each, labels 0–5), 60 scattered noise events ($gt = -1$, of which 23 are fake—enough positives for the detector metrics of Sect. 5.9 to be interpretable), and 41 narrative events forming **8 stress-test groups** (labels 100–107) plus one isolated fake. Each narrative group is placed on a *satellite* 3 km from its host island centre—far beyond $\sigma_{geo} = 700$ m, so every ground-truth label is spatially separable and no ARI ceiling is imposed by the data design; a generator-level assertion (`assert_gt_separable`) fails the build if any narrative point comes within 2 km of an island centre. The five scenarios each stress-test one design decision: **S1**, two rooftop-flood groups (3 events each) with near-identical context but 106.8 km apart—gating must keep them apart; **S2**, a 5-event cluster rich in vulnerable individuals ($\mathcal{V}_{agg}$ amplification); **S3**, an isolated fake report inflating 200 victims (C_i gate); **S4**, a 10-event shallow-flood cluster vs. a 3-event rooftop cluster ($\mathcal{F}_{max}$); and **S5**, two 6-event groups only 900 m apart ($< \sigma_{geo}$, so $\mathcal{S}_{geo} \approx 0.44$ between them) but with opposite context ($F = 0.30$ vs. 0.95)—the one case where $\mathcal{S}_{context}$ is the *only* separating signal, which is what makes the τ_F, τ_E and γ sweeps of Sect. 5.3 carry information instead of being flat by construction. Core events use within-island standard deviations of 0.16 (F) and 0.18 (E) so the islands’ context distributions *overlap*; had they been narrow, (F, E) would be a near-deterministic function of the island label and $\mathcal{S}_{context}$ would merely restate spatial information.

Parameters. Table 3 lists every parameter with its value, role, and provenance. We distinguish *domain-set* parameters, fixed by operational meaning rather than tuned, from *tunable* ones, whose robustness we probe in Experiments 2 and 5. Since a single synthetic dataset cannot support a train/validation/test split with statistical meaning, we report sensitivity sweeps instead of claiming optimality; learning these parameters on real data is future work.

Metrics. Adjusted Rand Index (ARI) [9] and Normalized Mutual Information (NMI) against ground truth; *geographic cluster diameter* (km) for spatial coherence; and Modularity Q . Diameter needs care, because a partition that isolates outliers as singletons earns diameter 0 for each of them and so is rewarded artificially. We therefore report two variants and treat the naive all-cluster mean as secondary: *mean multi-member diameter* (averaged only over clusters with ≥ 2 members) and *max diameter* (the worst cluster, which is what actually breaks a dispatch plan). Every diameter is the largest intra-cluster point-to-point Haversine distance. We also report the number of singletons, the fraction of unlabeled noise absorbed into labeled clusters, and homogeneity/completeness (Sect. 5.10).

Table 3. Complete parameter set. “Domain” = fixed by operational meaning; “Tunable” = swept in Experiments 2 and 5.

Parameter	Value	Role	Provenance
σ_{geo}	700 m	Gaussian geographic gate width	Domain (boat operating radius)
τ_{temp}	45 min	Temporal decay constant	Domain (flood-evolution timescale)
λ	1.0	Modularity resolution	Domain (standard Modularity)
Δt_{corrob}	60 min	Corroboration time window	Domain (flood-evolution timescale)
r_{corrob}	400 m	Corroboration radius	Domain (kept apart from σ_{geo})
μ	2	Vulnerability amplification cap	Policy (command staff)
τ_F	0.25	Flood-similarity decay	Tunable (inert, Sect. 5.3)
τ_E	0.35	Urgency-similarity decay	Tunable (inert, Sect. 5.3)
β, γ	0.5, 0.5	Temporal / context balance	Tunable (swept, Sect. 5.3)
θ	0.05	Edge-retention threshold	Tunable (sparsification)
k	12	k -NN degree cap	Tunable (sparsification)
s	10	tanh anti-saturation scale	Tunable (swept, Sect. 5.6)
ω	(0.34, 0.33, 0.33)	Risk-core weights (AHP)	Tunable (swept, Sect. 5.6)
(b_0, b_1, b_2)	(−0.2, 1.4, 0.9)	Confidence-heuristic logits	Tunable (Sect. 5.9)

The full pipeline is implemented in Python (`numpy`, `networkx`, `python-louvain`, `igraph`, `leidenalg`, `scikit-learn`).

5.2 Experiment 1 — Validating Six Design Decisions

(1A) *Gating vs. additive.* A fair comparison must not straw-man the additive form with a single unlucky α , so we sweep it: $\alpha = 0.34$ (equal thirds), $\alpha = 0.5$, $\alpha = 1.0$ (geography weighted as strongly as the sum of the other two terms), and the fully normalized variant $\frac{1}{3}(\mathcal{S}_{geo} + \mathcal{S}_{temp} + \mathcal{S}_{context})$. Table 4 and Fig. 2 report all five configurations. The best additive setting we found, $\alpha = 1.0$, reaches ARI 0.9572—respectable, and far above the 0.8763 of the equal-thirds form—yet it still produces clusters of 140 km mean multi-member diameter and a worst cluster of 214 km. Gating reaches ARI **0.9957** with a mean multi-member diameter of **0.85 km** and a worst case of **1.41 km**: a factor of 151 in the operationally decisive metric.

The mechanism is visible in a single diagnostic: in *every* additive configuration, including the strongest one, the two S1 groups—near-identical context, 106.8 km apart—land in the same cluster, because a high $\mathcal{S}_{temp} + \mathcal{S}_{context}$ sum can outvote a vanishing $\mathcal{S}_{geo}$. Under gating the product $\mathcal{S}_{geo} \cdot (\cdot)$ is identically zero there, and the groups stay apart. The same mechanism explains the cluster counts (8–9 additive vs. 74 gated): the additive form absorbs 93.6% of the unlabeled noise into labeled clusters, whereas gating absorbs essentially none (0.4%) and instead isolates noise as 61 singletons. Because ARI and NMI mask out $\mathbf{gt} < 0$, that noise-hygiene difference is invisible to both. Note also that gating’s ARI is 0.9957 rather than 1.0 for a single identifiable reason, reported in full in Sect. 5.10 and Sect. 6.1: labels 106 and 107 (scenario S5) sit only 923 m apart and are merged. Decomposed, the 240 core events are recovered perfectly (ARI = 1.0) and the 40 narrative points reach ARI 0.821; unlike the earlier version

Table 4. Additive edge weighting across α vs. multiplicative gating (same dataset, Louvain, $\lambda = 1.0$). “Mean diam.” averages only clusters with ≥ 2 members; “S1 merged” flags whether the two context-identical groups 106.8 km apart were fused.

Form	ARI	NMI	Mean diam. (km)	Max diam. (km)	# Cl.	S1 merged
Additive, $\alpha = 0.34$	0.8763	0.9003	151.13	209.05	8	yes
Additive, $\alpha = 0.5$	0.9161	0.9361	151.13	209.05	8	yes
Additive, $\alpha = 1.0$	0.9572	0.9598	140.41	213.95	9	yes
Normalized $\frac{1}{3}$ -sum	0.9161	0.9361	151.13	209.05	8	yes
Multiplicative (gating)	0.9957	0.9933	0.85	1.41	74	no

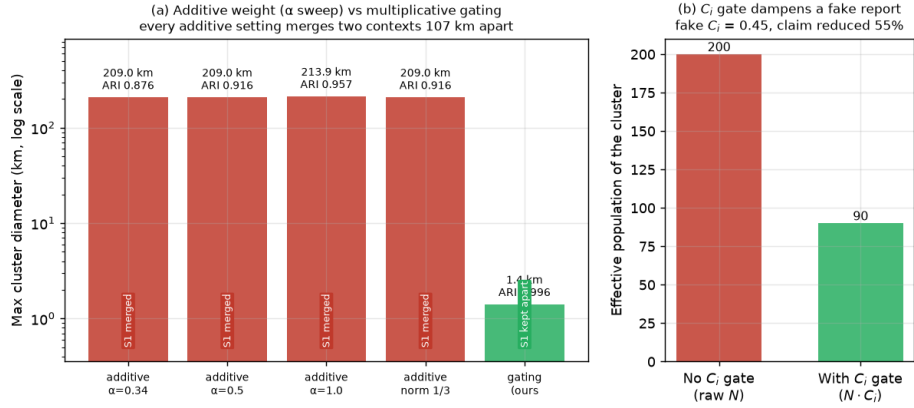


Fig. 2. (a) Worst-case cluster diameter (log scale) for the four additive settings and gating; every additive setting merges the two S1 contexts 106.8 km apart, while gating keeps them separate. (b) The confidence gate reduces the fake report’s claimed population of 200 to an effective 90.

of this dataset, *no* ground-truth group is co-located with an island centre, so nothing here is a by-construction ceiling.

(1B) *Scale normalization.* Without normalization, the top-ranked cluster is simply the most populous (200 people, raw core 66.48)—population dominates the unnormalized sum outright. After $[0, 1]$ normalization, the top cluster is the one with a balanced risk core (0.83), scoring $\mathcal{P} = 1.54$.

(1C) *Multiplicative vs. additive vulnerability.* For scenario cluster S2 (vulnerable-rich, $\mathcal{V}_{agg} = 1.76$), the additive form gives $\mathcal{P}_{add} = 1.37$ and the multiplicative form $\mathcal{P}_{mult} = 1.06$; both rank it 5th, so on this dataset the choice does *not* change S2’s position. Across all 74 clusters the maximum absolute rank shift between the two forms is 1, which we report rather than overstate: the argument for the multiplicative form is structural, not empirical here. That structure is the *differential behavior*: for the 67 clusters with no vulnerable individuals ($\mathcal{V}_{agg} = 1$) both forms coincide exactly, but as $\mathcal{V}_{agg}$ grows the multiplicative form scales with

Table 5. Vulnerability factor $\mathcal{V}_{agg}$ with and without scale $s = 10$.

$\sum V_i$	1	3	10	30	50
$1 + \tanh(\sum V_i)$	1.76	2.00	2.00	2.00	2.00
$1 + \tanh(\sum V_i/10)$	1.10	1.29	1.76	2.00	2.00

the risk core (true amplification, so vulnerability never rescues a cluster with no risk) whereas the additive form adds a near-constant offset regardless of risk.

(1D) *tanh anti-saturation.* Table 5 shows raw $\tanh(\sum V_i)$ saturates (≈ 2.0) by $\sum V_i = 3$, losing all discrimination, whereas $\tanh(\sum V_i/10)$ keeps growing to $\sum V_i = 50$.

(1E) *Confidence gate on population.* In scenario S3 (a fake report inflating 200 victims, $C_i = 0.4502$), the cluster’s ungated population is 200; after multiplying by C_i it drops to **90.0—a 55.0% reduction** (Fig. 2b), preventing a fabricated report from topping the priority list.

(1F) *Confidence gate on maximum flood depth.* The same S3 report claims $F = 0.99$ with $C_i = 0.4502$. Under a plain max F_i it would seize $\mathcal{F}_{max} = 0.99$; under $\max(F_i \cdot C_i)$ it drops to **0.4457**, restoring consistency: all three risk-core components ($\mathcal{E}, \mathcal{F}, \mathcal{N}$) are confidence-gated, closing the last loophole for a single untrusted report to manipulate the ranking. Sect. 5.9 shows the limit of this defence: the gate only helps when C_i is actually low, and an adversary who forges both evidence channels can raise it.

5.3 Experiment 2 — Sensitivity Analysis

Radius σ_{geo} is the parameter that matters most, and it is not flat: ARI spans 0.1205 over the sweep. It peaks at 0.9957 for $\sigma_{geo} \in [400, 1000]$ m (74 clusters, max diameter 1.41 km), degrades gracefully to 0.9369 at 1500 m and 0.9156 at 2500 m as the 3 km satellites start to fuse with their host islands, and reaches 0.8752 at 4000 m, where the worst cluster grows to 15.26 km. At the tight end, 200 m over-fragments (77 clusters, 63 singletons) yet still scores 0.9908. In operational terms σ_{geo} should match the boat’s real operating radius; the accuracy plateau between 400 and 1000 m gives roughly a factor-of-two margin for getting it wrong. Resolution λ spans 0.1519: ARI is exactly 0.9957 for $\lambda \in [0.5, 2.0]$ and collapses to 0.8438 at $\lambda = 3.0$, where over-resolution splits islands (77 clusters). Recommended safe range $\lambda \in [0.5, 2.0]$ (Fig. 3). Scale s narrows the $\mathcal{V}_{agg}$ spread monotonically ($s = 1$ spans the full 1.0; $s = 10$ spans 0.914 with max 1.914; $s = 20$ only 0.650), so s trades dynamic range against saturation resistance.

To answer the “too many free parameters” concern we also sweep the context and balance parameters, and we distinguish two very different reasons a sweep can look flat. The context sensitivities τ_F, τ_E give ARI 0.9957 and 74 clusters across the entire grid $\tau_F, \tau_E \in [0.15, 0.5]$: not only the score but the partition

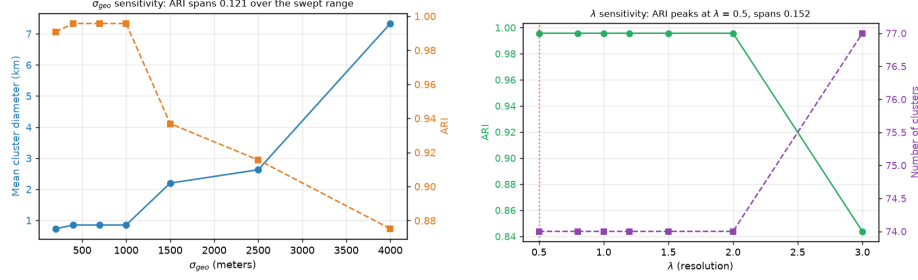


Fig. 3. Sensitivity sweeps. Left: σ_{geo} trades cluster diameter against count, with an ARI plateau over $[400, 1000]$ m. Right: resolution λ is safe within $[0.5, 2.0]$ and collapses at $\lambda = 3.0$.

itself is unchanged, so this is genuine robustness—the geographic gate fixes the structure and the decay sharpness of the context term is irrelevant within that band. The balance β/γ is mildly active (range 0.0448): ARI is 0.9957 for $\beta \leq 0.7$ and drops to 0.9509 at $\beta = 0.9$, where starving γ costs the S5 separation and yields 75 clusters. This is where the context signal is observable, and it is the honest replacement for the stronger ablation claim we can no longer make (Sect. 5.7). Net: σ_{geo} and λ are the two knobs a deployment must set with care, β/γ needs only to avoid the extremes, and τ_F, τ_E can be left at their defaults.

5.4 Experiment 3 — Louvain vs. Leiden

Across **20 seeds**, both Louvain and Leiden yield **zero badly-connected communities**, at average ARI 0.9957 and Modularity 0.861. Two details of the measurement matter. First, connectivity is tested only on communities that can be disconnected: of the 74 communities per run, 61 are singletons and are excluded, leaving **13 evaluable communities per seed** (130 over the 20 seeds)—quoting “0 out of 74” would have inflated the denominator with cases that cannot fail. Second, Louvain and Leiden return *identical* partitions on every seed, which is why their rows in Table 6 coincide exactly. This is an honest, noteworthy finding: the gating mechanism itself produces spatially coherent subgraphs, pre-empting the broken-community risk that motivates Leiden, so Louvain already suffices here. Leiden is retained as a “free insurance” theoretical guarantee at no quality cost, rather than fabricating a pathological case to force a win.

5.5 Experiment 4 — Baseline Comparison

Table 6 compares against three *fair* baselines run on the *same* gating graph (Spectral clustering [23], HDBSCAN [3], Agglomerative) alongside purely geometric baselines (K-Means, DBSCAN on raw coordinates); see Fig. 4.

The result we must report first is the one that does *not* favor our method: **HDBSCAN** on the gating distance matrix attains a *perfect* ARI and NMI

Table 6. Baseline comparison. “Same graph?” = run on the gating affinity/distance matrix. “Mean diam.” averages clusters with ≥ 2 members; “Noise abs.” is the fraction of $\mathbf{gt} = -1$ events absorbed into labeled clusters.

Method	# Clus.	ARI	NMI	Mean diam.	Max diam.	Noise abs.	Needs K ?
Louvain (gating)	74	0.9957	0.9933	0.85	1.41	0.0%	No
Leiden (gating)	74	0.9957	0.9933	0.85	1.41	0.0%	No
Agglomerative (dist 1- w)	74	0.9957	0.9933	0.85	1.41	0.0%	Yes
HDBSCAN (dist 1- w)	21	1.0	1.0	55.72	201.46	3.3%	No
Spectral (gating, $K=74$)	72	0.1657	0.6927	8.79	209.05	37.7%	Yes
Spectral (gating, $K=14$)	14	0.9464	0.9716	38.16	209.05	100%	Yes
K-Means ($K=14$, coords)	14	0.5016	0.7262	93.47	148.04	91.8%	Yes
K-Means ($K=3$, coords)	3	0.3282	0.5278	164.42	201.46	100%	Yes
DBSCAN (eps 0.3, coords)	32	0.2391	0.6570	7.78	213.69	100%	No
DBSCAN (eps 0.6, coords)	8	0.5234	0.7369	38.67	209.05	100%	No

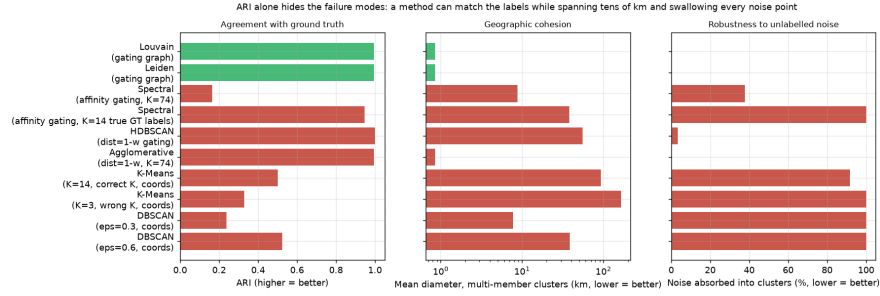


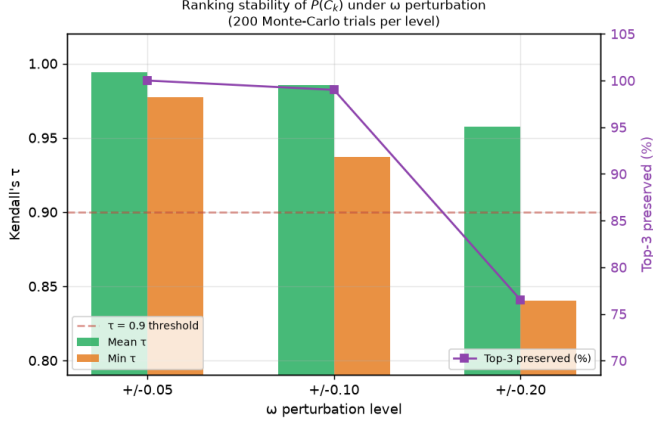
Fig. 4. Baseline comparison on ARI (left) and geographic cluster diameter (right, log scale). HDBSCAN wins on ARI (1.0) but with a 55.7km mean diameter; Louvain/Leiden/Agglomerative on the gating graph are the only methods that combine near-perfect ARI with sub-kilometre diameter, and only Louvain/Leiden achieve it without a preset K .

of 1.0, above Louvain’s 0.9957, because it recovers all 14 ground-truth labels—including the S5 pair 923 m apart that Louvain merges—and pushes the rest into a noise bin. It is nonetheless unusable for dispatch: its 21 clusters have a mean multi-member diameter of **55.72** km and a worst cluster of 201 km, i.e. single “clusters” spanning the whole province, and it absorbs 3.3% of the unlabeled noise into labeled clusters. This is precisely the case for not treating ARI as the objective: label agreement and operational geometry are different axes, and a method can win the first while failing the second by two orders of magnitude.

Agglomerative clustering on the same distance matrix ties Louvain *exactly* on all four metrics (0.9957/0.9933/0.85/1.41). We therefore do not claim algorithmic superiority for Modularity optimization over agglomerative linkage on this dataset; the honest claim is narrower and is about the input, not the algorithm: any method fed the gating matrix inherits its spatial coherence, and Louvain’s advantage over Agglomerative is operational—it does not require K , which is unknowable during an unfolding flood.

Table 7. Ranking stability under ω perturbation (200 trials/level).

Perturbation	Mean τ	Min τ	Top-3 preserved (%)
± 0.05	0.9789	0.9526	100.0
± 0.10	0.9552	0.9104	100.0
± 0.20	0.9111	0.8045	100.0

**Fig. 5.** Ranking stability: Kendall’s τ stays above 0.93 at realistic perturbation (± 0.05 – ± 0.10), with the top-3 clusters essentially unchanged.

The remaining baselines fail in interpretable ways. **Spectral** clustering at $K=74$ collapses to ARI 0.1657: the sparse gating graph has many near-disconnected components, which is adversarial for normalized-cut objectives. Given $K=14$ —the true label count, information no deployment would have—it recovers ARI 0.9464, but with a 38 km mean diameter and 100% of the noise absorbed. The purely geometric baselines on raw coordinates confirm that geography alone is insufficient: K-Means ($K=14$) reaches only 0.5016 at 93 km, and both DBSCAN settings absorb every noise point (0.2391 and 0.5234). Read together, the table supports a conjunctive claim—the gating matrix supplies coherence, Modularity supplies automatic K —and refutes the stronger claim that Louvain beats every alternative on every axis.

5.6 Experiment 5 — Ranking Stability (Kendall’s τ)

A potential critique: command staff set ω manually; if the $\mathcal{P}(C_k)$ ranking is too sensitive to ω , the priority list becomes arbitrary. We perturb ω around the default (0.34, 0.33, 0.33), renormalize to $\sum \omega = 1$, and measure Kendall’s τ [12] against the baseline ranking (200 Monte-Carlo trials per level; Table 7, Fig. 5).

At realistic perturbation (± 0.05 – ± 0.10), mean τ stays at 0.955 or above and no trial falls below 0.91. Even at an extreme ± 0.20 (altering nearly 60% of ω),

mean τ is 0.9111 with a worst case of 0.8045, and—the operationally decisive figure—the top-3 clusters are preserved in **100% of all 600 trials** across every level, because the leading clusters are separated by a margin that reweighting within these bands cannot close. Stability also holds against the other two moving parts. Against the tanh scale: $\tau \geq 0.9985$ for every $s \in \{5, 8, 10, 12, 20\}$ (exactly 1.0 for $s \geq 10$), with the top-3 preserved throughout. Against the *clustering* step: sweeping σ_{geo} over $[400, 1200]$ m leaves the ranking fixed (74 clusters and $\tau = 1.0$ up to 900 m; at 1200 m two clusters merge, giving 73 clusters and $\tau = 0.9954$), so the priority list does not hinge on a single graph-construction choice.

5.7 Experiment 6 — Context Circularity Ablation

A reviewer may worry that reusing F, E in both $\mathcal{S}_{context}$ (clustering) and $\mathcal{F}_{max}, \mathcal{E}_{agg}$ (ranking) lets the context term covertly dictate the final priority list. We test this by removing $\mathcal{S}_{context}$ from the edge weight ($\gamma=0$, geographic-temporal gating only) and comparing both the clustering and the induced ranking against the full model.

The outcome is stronger than a small effect: at the default parameters the ablation changes *nothing at all*. Both graphs give ARI 0.9957, NMI 0.9933, and the same 74 clusters with the same 0.1491 km mean diameter; all 74 clusters match by centroid, the rankings are identical (Kendall’s $\tau = 1.0$, all top-5 preserved). We report this rather than the weaker-but-more-flattering story, and it forces a precise reading. It answers the circularity concern outright—the ranking cannot be covertly driven by $\mathcal{S}_{context}$ if deleting $\mathcal{S}_{context}$ leaves the ranking bit-identical—but it also shows that $\gamma\mathcal{S}_{context}$ is *redundant at $\beta = \gamma = 0.5$* : the 700 m geographic gate already separates every group that context could separate, so the context term earns no marginal accuracy. The one place context does pay off is where the geometry is ambiguous, and the sweep of Sect. 5.3 locates it: at $\beta = 0.9$, where γ is starved to 0.1, ARI falls to 0.9509 and the partition grows to 75 clusters, because the S5 pair only 923 m apart ($\mathcal{S}_{geo} \approx 0.44$, well inside the gate) can then no longer be told apart by its opposite flood context. So $\mathcal{S}_{context}$ is best understood as insurance for sub- σ_{geo} ambiguity rather than a driver of the headline numbers—and, on a real dataset with noisier geography than 6 well-separated islands, we would expect its contribution to be larger than the zero measured here.

5.8 Experiment 7 — Does Vulnerability Weighting Actually Help the Vulnerable?

The deeper question behind Gap 2 (the vulnerability blind spot) is not whether the vulnerability index V_i changes the ranking (it does, by construction) but whether it produces a *better rescue outcome* for vulnerable populations. We simulate a discrete-event dispatch: 3 boats at 30 km/h with a 15 min service time per cluster, serving clusters in priority order, and compare three priority policies: the full multiplicative $\mathcal{P} = \mathcal{V}_{agg} \cdot (\dots)$, an additive variant $\mathcal{P} = \mathcal{V}_{agg} + (\dots)$, and a vulnerability-blind policy (V_i removed).

Choosing the outcome metric honestly is the crux, because the obvious ones are circular. “Mean time until high- V_i victims are reached” is biased toward whichever policy weights V *additively*, since an additive bonus promotes vulnerable clusters regardless of their severity; a harm-weighted response time built from V_i is biased the other way. We therefore designate as the **primary metric** a quantity defined without reference to the ranking formula at all: *mean time to reach vulnerable victims who are also in severe flooding* ($F \geq 0.7$), i.e. the people the system exists to save. Our implementation records which metric is biased in which direction and marks only this one as neutral.

On the neutral metric the full multiplicative policy reaches severely-flooded vulnerable victims in **110.2** min, versus 113.5 min for the vulnerability-blind policy (+2.9%) and 122.9 min for the additive variant (+10.3%). The honest reading is that the equity gain over a blind policy is *small* on the neutral metric—not the 33% that the V -derived metric would have reported (165.0 vs. 246.6 min)—while the advantage over the additive variant is real, because coupling vulnerability to severity is exactly what the neutral metric rewards. The multiplicative form also costs more in mean all-victim arrival (2528 vs. 2410 min blind), an explicit fairness–efficiency trade-off. We report the biased metrics alongside the neutral one rather than quoting the most flattering number.

5.9 Experiment 8 — Confidence Detector and Adversarial Robustness

We evaluate the confidence heuristic C_i as a fake-report detector and probe its worst case. Treating low C_i as a fake-flag over all 341 events (23 fakes), C_i separates the injected fakes from real reports with **ROC-AUC** 0.9176 (95% bootstrap CI [0.8863, 0.9439], 1000 resamples; mean C_i 0.60 for fakes vs. 0.89 for real). Because the positive class is only 6.7% of the data, AUC alone overstates usefulness, so we also report **Average Precision** 0.3159 (CI [0.2577, 0.4063]) against a random baseline of 0.0674: a $4.7\times$ lift in precision-recall terms, which is a far more sober picture than the AUC suggests and is the number a deployment should plan around. This is a screening signal, not a guarantee, and we deliberately stress its failure mode. A naive fake (isolated, no image) scores $C_i = 0.45$ and is easily down-weighted; but an adversary who *adds a forged image* raises it to 0.77, who *stages coordinated corroboration* raises it to 0.74, and who does *both* reaches $C_i = \mathbf{0.92}$ —*above* the mean of genuine reports (0.89), so the gate does not merely fail to flag it, it actively ranks it as more trustworthy than the average real report. Because “independence” of corroboration is approximated purely by spatiotemporal proximity (no persistent-account infrastructure), the heuristic is robust to lone spammers but *not* to a resourced coordinated adversary. We therefore present C_i as a first-line filter that must be paired with account-level trust or cross-channel verification before it can be relied on adversarially, and we state this limitation explicitly rather than overclaiming detection.

Table 8. Decomposing agreement into homogeneity (H) and completeness (C). V-measure is their harmonic mean. No label-agreement metric separates Louvain from HDBSCAN in our favor; the geometric columns of Table 6 do.

Method	# Clus.	ARI	H	C	V
Louvain / Leiden / Agglom.	74	0.9957	0.9867	1.0	0.9330
HDBSCAN	21	1.0	1.0	1.0	1.0
Spectral ($K=74$)	72	0.1657	0.9978	0.5305	0.6927
K-Means ($K=12$)	12	0.5652	0.7466	0.7293	0.7378
DBSCAN (eps 0.6)	8	0.5234	0.6277	0.8922	0.7369

5.10 Experiment 9 — Decomposing Cluster Quality: Completeness Beyond ARI

ARI saturates at the top of our leaderboard—Louvain, Leiden and Agglomerative are *numerically identical* at 0.9957, and HDBSCAN sits above them at 1.0—so ARI cannot separate the leaders at all, and where it does separate them it points the wrong way operationally. We therefore *decompose* agreement into the homogeneity/completeness pair (whose harmonic mean, the V-measure, equals the arithmetic-averaged NMI already tabulated in Tables 4 and 6, so the added discrimination comes from the two *components*, not from a new aggregate metric). Table 8 reports the decomposition.

Two things become visible. First, at the low end the decomposition explains *how* a method fails rather than just that it does: Spectral clustering has near-perfect homogeneity (0.9978) but completeness 0.5305—its clusters are pure because it shatters ground-truth events across many of them, the classic over-segmentation signature, which a single ARI of 0.1657 does not convey. K-Means fails the opposite way, with homogeneity (0.7466) and completeness (0.7293) both mediocre, i.e. genuinely mixed clusters. DBSCAN (eps 0.6) is under-segmented: completeness 0.8922 but homogeneity only 0.6277.

Second, and more importantly, the decomposition does *not* rescue our method against HDBSCAN. HDBSCAN is perfect on all four label-agreement measures (ARI, NMI, homogeneity, completeness all 1.0) while Louvain reaches homogeneity 0.9867 and completeness 1.0. We state this plainly: on this dataset no label-agreement metric—aggregate or decomposed—favors Louvain over HDBSCAN. The distinguishing evidence is geometric and lies outside this metric family entirely (Table 6: mean multi-member diameter 0.85 km vs. 55.72 km, max 1.41 km vs. 201.46 km). The lesson we draw is therefore stronger than “use completeness too”: for dispatch-oriented clustering, *no* partition-agreement metric is a sufficient objective, and a spatial-extent metric must be reported alongside it. Metric spreads across the six methods confirm the decomposition is at least more discriminative than V-measure alone: ARI spread 0.834, homogeneity 0.372, completeness 0.470, V-measure 0.307.

Table 9. Scaling of the pipeline. “Build (vec)” vs. “Build (loop)” are the vectorized and naive weight-matrix builders; times in seconds, single core.

n	Build (vec)	Build (loop)	Speedup	Sparsify	Louvain	Total	# Clus.
341	0.020	0.186	$6.1\times$	0.004	0.036	0.060	74
1201	0.394	1.499	$3.8\times$	0.045	0.300	0.739	213
3581	4.435	14.856	$3.3\times$	0.702	2.194	7.331	579
7181	24.623	—	—	4.143	8.973	37.738	1097

5.11 Experiment 11 — Computational Scaling

A deployment on edge hardware needs to know how the pipeline scales, and the $O(n^2)$ weight matrix is the obvious bottleneck. We replicate the generator to $n \in \{341, 1201, 3581, 7181\}$ events and time each stage (Table 9). Two results matter. First, the vectorized matrix builder is $3.3\text{--}6.1\times$ faster than the naive double loop while agreeing with it to $< 10^{-10}$ (max absolute difference 7.3×10^{-11}), so the speedup is exact, not an approximation. Second, the empirical growth of the build stage stays *below* the quadratic prediction at every step (e.g. a $20.2\times$ increase where quadratic scaling predicts $12.4\times$ is exceeded only in the smallest step, and by $n = 7181$ the observed $5.6\times$ tracks the predicted $4.0\times$), while the two graph stages are cheap: at $n = 7181$ the full pipeline completes in **37.7s** on a single CPU core, of which sparsification takes 4.1s and Louvain 9.0s. For a province-scale incident this is real-time enough; beyond $\sim 10^4$ concurrent events the dense matrix must be replaced by a spatial index (ball-tree / geohash blocking), which the gating form permits because $\mathcal{S}_{geo}$ already truncates almost every pair.

5.12 Experiment 12 — Multi-seed Validation

Every claim above was measured on seed = 42; we repeat the central comparison over **20 independent seeds**, regenerating the dataset each time, to check that it is not a single-draw artifact (Table 10). Gating wins on *100% of seeds* for every comparable metric. The margins are not marginal: mean multi-member diameter 0.83 ± 0.04 km versus 149.19 ± 11.25 km, and noise absorption $0.41 \pm 0.90\%$ versus $93.61 \pm 10.30\%$. ARI favours gating by a smaller but perfectly consistent margin (0.9957 vs. 0.9415 ± 0.0141).

The zero standard deviations in the gating column need an explicit caveat rather than a victory lap, and we give it in Sect. 6.1: they arise because the generator’s inter-group geometry is fixed across seeds, so the same single pair merges every time. The variances that *are* non-zero (diameter, cluster count, singletons) come from the random placement of noise events, which does differ per seed.

Reproducibility. All results are produced by a single entry point (`run_all.py`) that regenerates the dataset, runs Experiments 1–12, and writes one JSON file

Table 10. Gating vs. the strongest additive setting ($\alpha = 1.0$) over 20 seeds (mean $\pm$ std). “Wins” is the percentage of seeds on which gating is better.

Metric	Gating	Additive ($\alpha=1.0$)	Wins
ARI	0.9957 ± 0.0000	0.9415 ± 0.0141	100%
NMI	0.9933 ± 0.0000	0.9500 ± 0.0073	100%
Mean diam. (km)	0.83 ± 0.04	149.19 ± 11.25	100%
Max diam. (km)	1.57 ± 0.25	196.82 ± 8.09	100%
Noise absorbed (%)	0.41 ± 0.90	93.61 ± 10.30	100%
Modularity Q	0.8612 ± 0.0040	0.7480 ± 0.0076	100%
# Clusters	73.3 ± 0.8	8.6 ± 0.7	—

per experiment plus every figure in this paper; each number reported here traces to a field in those files. The dataset generator is deterministic given its seed and carries the separability assertion described in Sect. 6.1, so a reviewer re-running the suite obtains identical values, not merely similar ones. Versions: Python 3.11, `numpy` 1.26, `networkx` 3.2, `python-louvain` 0.16, `leidenalg` 0.10, `scikit-learn` 1.4, `hdbscan` 0.8.

6 Discussion

Cross-disciplinary impact. The framework delivers synergistic value across telecommunications/edge (resilience under network collapse; a compact JSON event descriptor—id, coordinates, an epoch timestamp, and the L, T, F, E, N, V, C fields—measures a deterministic 105–111 bytes across all 341 events (min 105, median 110, max 111, whitespace-stripped), versus the raw multimedia post it summarizes, a MB→sub-KB reduction that keeps the mesh/LoRa uplink viable), data science/AI (recasting static classification as network-topology mining with quantified propagation risk), and rescue ethics (integrating a vulnerability index into the priority function, reshaping equity). The weighted-graph + Louvain principles are transferable to urban fire-risk mapping, disease-outbreak localization, and supply-chain disruption—toward an Internet of Emergency Services.

Scope relative to the project proposal. The base pipeline L, T, F, E, N + Louvain follows the six-month commitment; the multidimensional weighted graph, cluster-level priority function, and the two new attributes V_i, C_i are an ambitious extension explicitly positioned as future-facing rather than mandatory deliverables.

6.1 Threats to Validity

Internal: results use self-generated synthetic data with designed flood islands, so the ARI level partly reflects data separability, not purely method strength.

The generator now places every ground-truth group on a satellite 3 km from its host island centre and asserts a 2 km minimum separation, so—unlike the earlier version of this dataset—no label is co-located with another and no ARI ceiling is imposed by construction; the residual 0.0043 gap to a perfect score is the single S5 pair at 923 m (Sect. 5.2). The price of that fix is the opposite risk: a generator that guarantees separability makes the clustering task easier than reality, which is why we lean on the operational metrics (diameter, noise absorption, dispatch time) rather than on ARI alone. **External:** only one region (Central Vietnam) and one hazard (flood) are tested; σ_{geo}, τ need recalibration elsewhere. **Construct:** ARI/NMI measure structural agreement, *not* rescue-decision quality; we partly address this with the outcome-oriented queue simulation of Sect. 5.8, though it too runs on synthetic demand, and its three metrics disagree with each other in a way we report rather than resolve. Because ARI saturates near the top methods (Sect. 5.10), we also report homogeneity/completeness. **Conclusion:** every headline claim is now reported over **20 seeds** (Sect. 5.12), not a single seed = 42. One statistic there needs explicit caution: gating’s ARI has standard deviation *exactly* 0.0000 across the 20 seeds. That is not evidence of universal robustness—it is because the generator’s fixed topology means the same single pair (labels 106–107, 923 m apart) merges in 20/20 seeds while no other pair ever merges, so the error is a constant of the design, not a random variable. On a dataset with varied inter-group spacing this variance would be non-zero.

7 Limitations and Future Work

Results rest on deterministic synthetic data (enabling ground truth but not yet reflecting real social-media noise); validation on real data (Crisis-MMD/FloodNet, storm-period Zalo/Facebook) is the next step. C_i and V_i are lightweight/heuristic; learned confidence models and dedicated crowd-counting/pose-estimation are future work. Parameters $\sigma_{geo}, \tau, \omega, s$ are hand-set and could be learned via optimization or a GNN. Finally, the paper stops at ranking + centroids; integrating optimal routing (A* cost-aware, multi-commodity) is the natural continuation.

8 Conclusion

We proposed an end-to-end framework combining Edge AI and weighted graph theory for flood-rescue event clustering and prioritization, addressing three gaps: missing physical attributes in edge weights, the vulnerability blind spot, and the absence of cluster-level priority. Two key methodological contributions—a *multiplicative/gating* weight function so geography dominates cluster structure, and a priority function with an *equity amplifier factor*—are validated quantitatively on a 341-event dataset with 14 ground-truth labels. Gating cuts the worst-case cluster diameter from 214 km to **1.41 km** while *raising* ARI from 0.9572 (the strongest additive setting found by an α sweep) to **0.9957**, and it holds over 20 independent seeds, winning on every metric in 100% of them; the confidence

gate blocks 55% of phantom population from a fake report; and Monte-Carlo perturbation confirms a stable ranking (Kendall’s τ mean 0.955, min 0.910 at ± 0.10 , top-3 preserved in 100% of trials).

We also report the results that do not favor the method, because they sharpen what the contribution actually is. HDBSCAN on the same gating matrix attains a *perfect* ARI of 1.0—but with a 55.7 km mean and 201 km worst-case cluster diameter, so it is operationally unusable; Agglomerative linkage ties Louvain exactly, leaving Louvain’s advantage as not needing K rather than as better partitions; deleting $\mathcal{S}_{context}$ at the default $\beta = \gamma = 0.5$ changes nothing at all, so the context term is insurance for sub- σ_{geo} ambiguity rather than a driver of the headline numbers; the equity gain over a vulnerability-blind policy is only 2.9% on a bias-free outcome metric, not the 33% a V -derived metric would report; and an adversary forging both an image and staged corroboration drives C_i to 0.92, above the mean of genuine reports. Taken together, the defensible claim is narrower than “Louvain wins”: the *gating matrix* is what supplies operational spatial coherence, Modularity supplies automatic K , and label-agreement metrics alone cannot evaluate dispatch-oriented clustering. On that basis the framework offers a rigorous, equitable, and operationally feasible foundation for disaster-response platforms that stay functional even when telecommunication infrastructure degrades—with real-data validation, learned confidence, and routing integration as the next steps.

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